

The whole thesis as one story

What happens in a shipyard, what the research asks, what has been done, what is under way and what comes next.

Road-occupying transport in shipyards: network-bound capacity, deadlock-free orchestration and fleet-network co-design

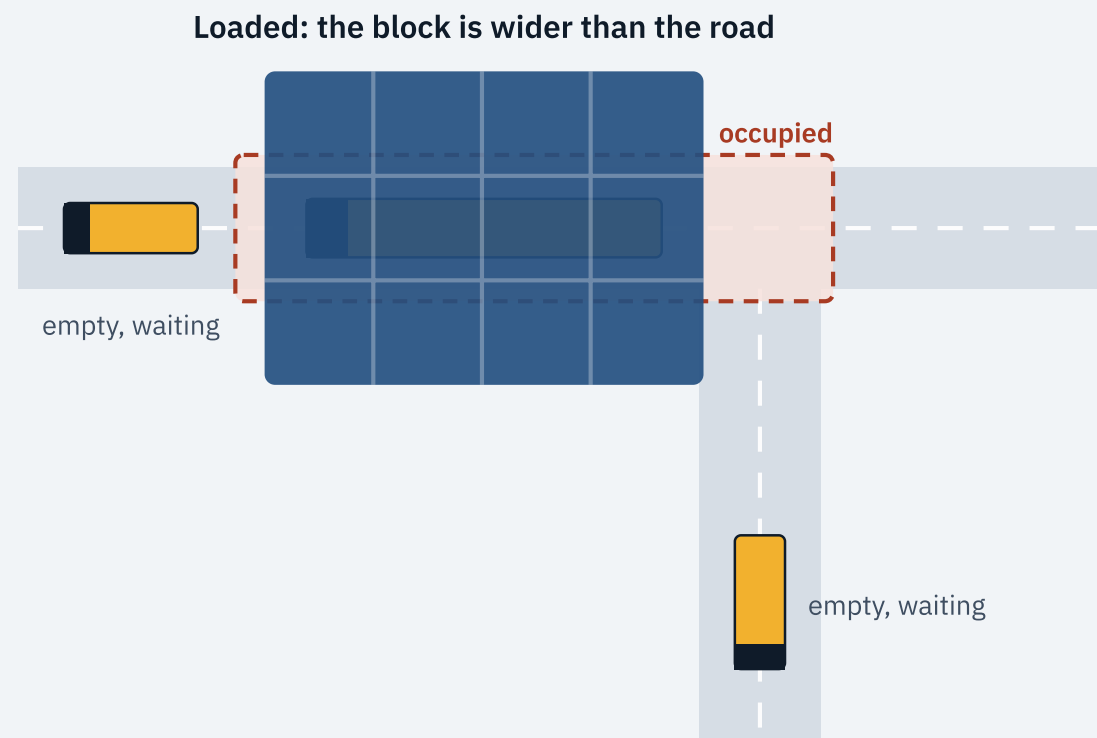
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As of 10 Oct 2026 · full results in the thesis overview

The usual model: vehicles never meet

Reality: a loaded vehicle owns the road



The order of this briefing (click to jump)

1

What happens in a shipyard

Blocks are wider than roads: a loaded vehicle takes the road

p. 3

2

What the research asks

Where the usual model goes wrong; the question, five claims, the capacity interval

p. 5

3

What has been done

New direction, flag experiment, two yards, three corrections, the dock mouth

p. 8

4

What is under way

Where deadlock comes from; a smarter rule that never jams

p. 14

5

What comes next

Vehicles or roads; more yards; theorems and the first paper

p. 16

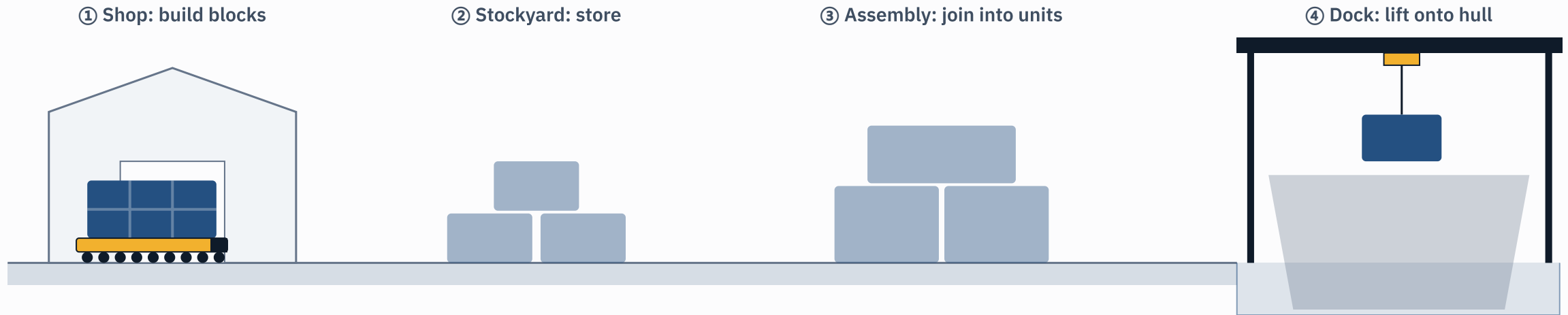
6

The whole story in one figure

What each chapter does on that figure

p. 17

A ship is built like a stack of blocks: blocks, transporters, docks and cranes



1

Blocks

The hull is built in pieces called blocks: hundreds of tonnes each, often 15–40 m long and wide.

2

Transporters

Hydraulic platform transporters carry each block between shops, stockyards and assembly areas several times. An empty one is about 7 m wide.

3

Docks and cranes

Finally a block goes to the dock, where a gantry crane lifts it onto the hull. When the cranes are busy, vehicles queue at the dock.

4

The key fact

Once loaded, vehicle and block are wider than the road: the whole segment and junction are taken, and oncoming vehicles must wait.

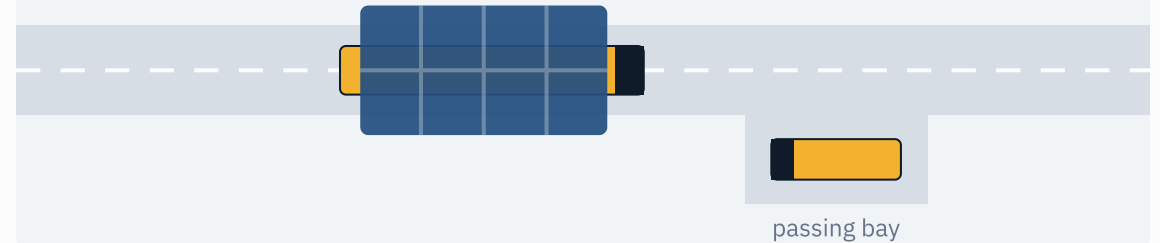
The usual model treats vehicles as points that never meet

Free-flow model: the two pass through each other



Travel times from a distance table, no constraint between vehicles

Reality: the empty one waits in a bay until the road is free



Earlier studies: free flow

Vehicles never affect each other; more vehicles always move more, at most with shrinking gains. Like cars on an empty highway.

Reality: occupancy

A loaded vehicle holds the segment and the junction alone; oncoming vehicles wait, with no passing and no overtaking. More like a train on a single track.

Researchers in the field have named this gap themselves: Sun et al. (2024) write in their conclusion that some roads take only one transporter at a time, and leave it as future work. In 288 delay records from a Shanghai yard, transporters occupy several lanes and other vehicles must give way (Wang et al. 2024).

Change the premise and four things change with it

1

The ceiling is the network

There is only so much road; extra vehicles just queue. How much moves each day is set, in the end, by the network.

2

More vehicles can be slower

Extra vehicles waiting on the road block everyone else.

3

Dispatching becomes traffic ordering

Scheduling shifts from handing out jobs to deciding who passes which road first, as in railway dispatching.

4

A road change can beat a vehicle

One passing bay, or splitting a road into sections, may be worth several vehicles.

In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.

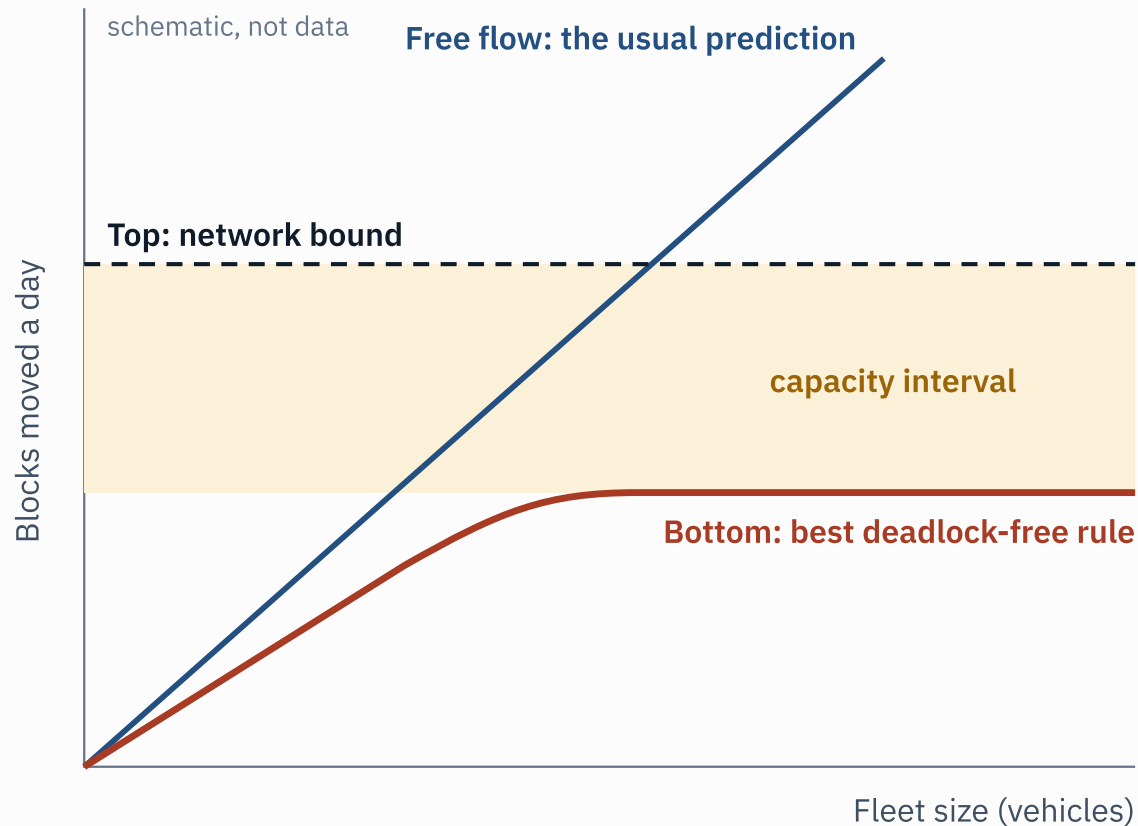
The question, and five bold answers

The question: when loaded vehicles own the road, what decides how many blocks a yard can move a day, and how should buying vehicles, scheduling and changing roads be designed together?

CLAIM	WHAT IT SAYS
C1 Two ceilings	Daily output is the smaller of what the fleet can run and what the network can let through; the second can be computed by linear programming.
C2 Trapezoid law	As vehicles are added, output first rises in a straight line, then levels off; if the extra vehicles wait on the road, it can even fall.
C3 Traffic ordering	Once output levels off, how much moves depends mostly on release order, where vehicles wait and how road segments are booked.
C4 Exchange rate	When the plateau nears the ceiling, how many vehicles a key road change is worth can be computed.
C5 It carries over	These patterns hold across yards and can be predicted from the map.

Seven hypotheses (H1–H7) were written down in advance, each with a result either way: if one holds we gain a finding, if not a useful boundary.

Capacity is an interval: the network sets the top, scheduling the bottom



Top: the network bound

The most the road network itself can let through, whatever scheduling rule is used.

Bottom: the best rule that never jams

What the best rule found so far achieves while guaranteeing that vehicles never lock each other in.

The gap says what to do

Bottom far below the top: improve scheduling first (chapter 4).

Bottom close to the top: only road changes help (chapter 5).

Step 1: a new direction that starts from a premise (7–8 October 2026)

1

The earlier topic

Two questions: can a block get from here to there at all (geometric feasibility), and which vehicles to buy, and how many (fleet sizing).

2

New conditions (7 Oct)

Entirely new content within shipyard transport scheduling; public data only, no precise annotation; at the research front.

3

Weighing candidates

Several directions, such as a digital-twin base or teams of vehicles carrying one block, each checked against the literature.

4

Road-occupying transport (8 Oct)

It questions a premise the whole field shares, not one paper. The same day set the rule that prior work is no veto.

The earlier papers: the remote-sensing segmentation paper is published (Ocean Engineering 2026); the fleet-sizing paper is under review and supplies algorithms and price data; the geometric-feasibility study is paused, to be decided as a separate paper.

Step 2: the flag experiment shows the effect is real (Okpo yard)

On the real road network of Hanwha Ocean's Okpo yard (centrelines traced from public imagery), each road segment and junction is a resource for one vehicle at a time, and a whole day of transport is simulated. Three rules for conflicts are compared:

Free flow

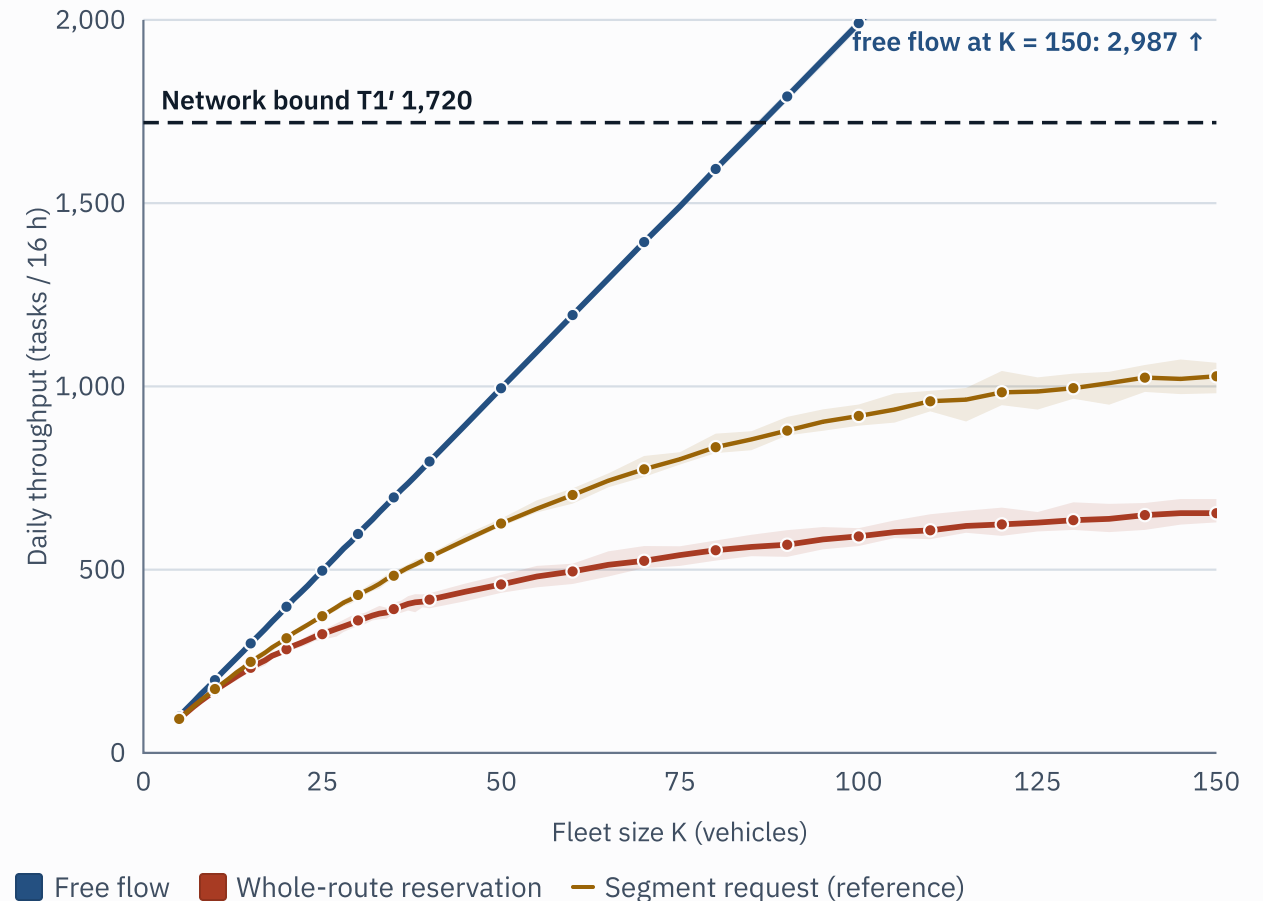
Vehicles never interact: the usual model.

Whole-route reservation

Book the entire route before leaving, as a train is given its path. It never jams, but it is cautious.

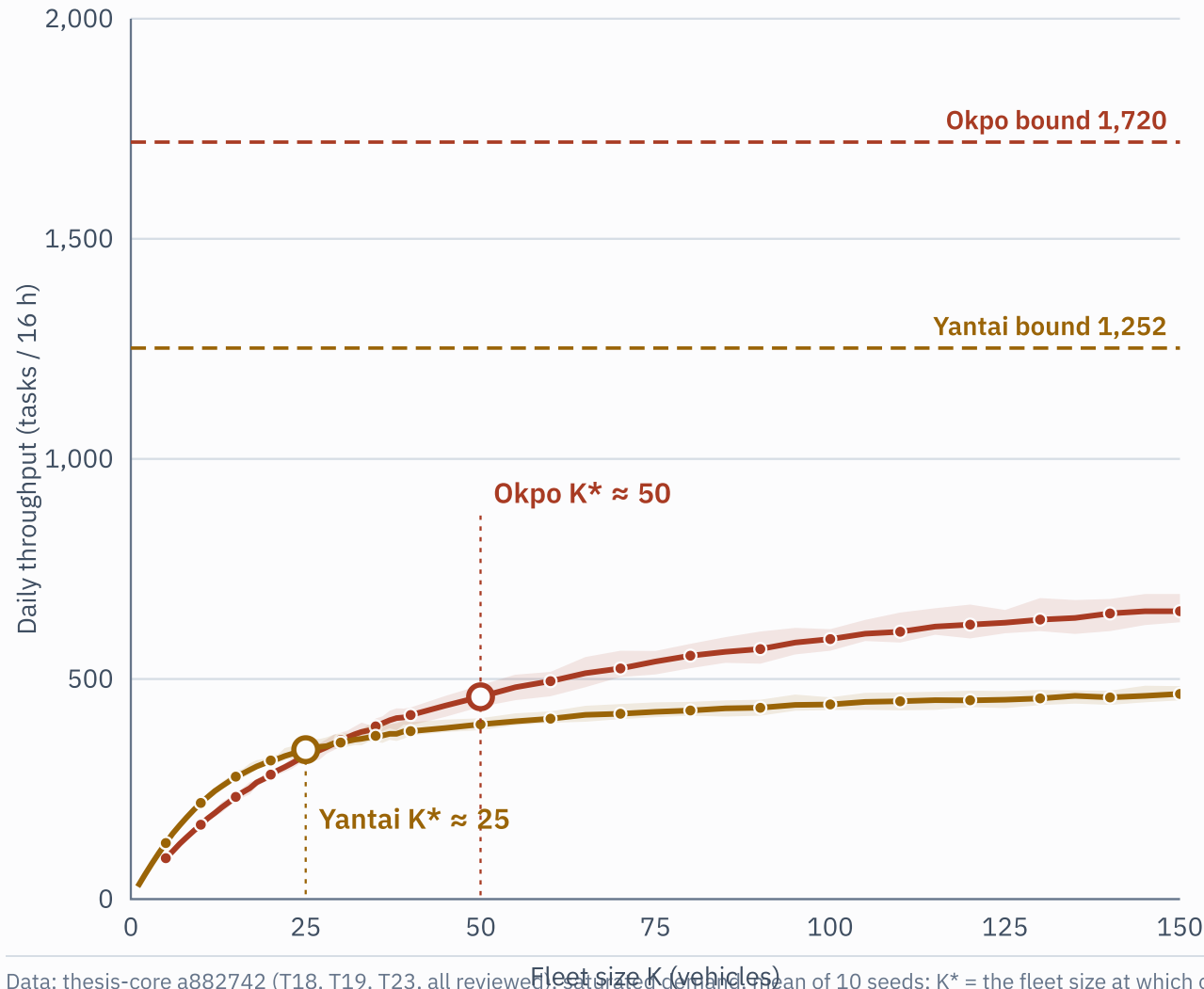
Segment request

Ask for one segment at a time: more efficient, but two vehicles can lock each other in a deadlock. Deadlocks were cleared by teleporting a vehicle out, so this rule is a reference only.



At K = 80: free flow 1,593, reservation only 553, 35% of it

Step 3: the ceiling, a second yard, the dock mouth (8–9 October)



Data: thesis-core a882742 (T18, T19, T23, all reviewed), saturated demand, mean of 10 seeds; K^* = the fleet size at which one more vehicle first adds less than 20% of its free-flow gain. The system bound without sampling noise is 1,736 for Okpo and 1,298 for Yantai.

■ Okpo (reservation) ■ Yantai (reservation) — Network bound T1' (dashed)

1

Pinning down the ceiling

Tightened network bound (dashed): about 1,720 a day at Okpo, 1,250 at Yantai; none of 50,000+ simulated points exceeds it. Each road also gets a shadow price.

2

A second yard: Yantai

Both level off, but network limits begin at different fleet sizes (dots): about 50 vehicles at Okpo, 25 at Yantai. Same pattern, layout-specific numbers.

3

Cranes at the dock mouth

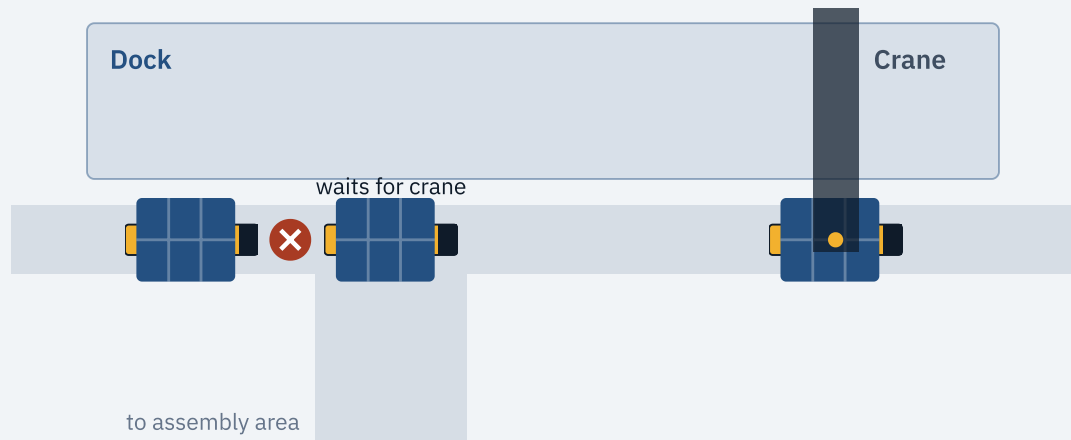
Busier cranes, longer dock queues; past a threshold, yard-wide capacity drops sharply. Step 4 shows why.

Step 4: the author reviews everything and three corrections follow (9–10 October)

PROBLEM	THE FIX	WHAT CAME OUT
Task volume too small: 97 or 194 a day was a four-vehicle subset	Whole-yard volume converted from public steel use	Okpo moves about 339 blocks a day, within 3% of the 350 Hanwha has published
The sharp drop at the dock relied on teleporting, which is not physical	A safe release rule that never teleports and never deadlocks (like a bank that lends only when every borrower can still finish); the dock road split at its stops	It still drops without teleporting; it does not drop when vehicles wait off the road; splitting the road moves the threshold later. The drop is real, and it is about where vehicles wait
Capacity should be an interval, not one number	[best deadlock-free rule, network bound]	Okpo [627, 1,720]: the bottom is about 36% of the top; scheduling decides the gap, which chapter 4 aims to win back

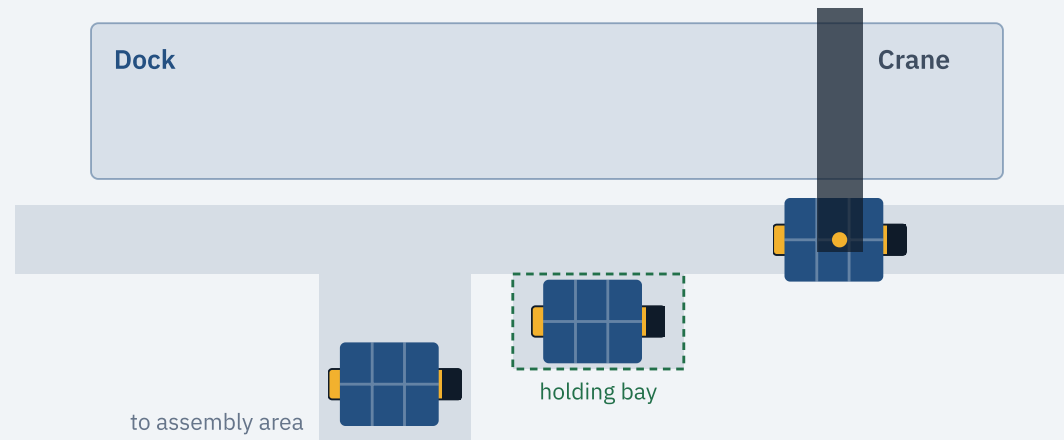
The dock mouth: where vehicles wait decides whether the yard is held up

Waiting for the crane on the dock road



The vehicle waiting for the crane holds the dock road, so traffic to the assembly area is stuck too and the queue spreads yard-wide. The same holds without teleporting.

Waiting in an off-road holding bay



The waiting vehicle stands off the road, so the way stays open and the queue stays at the dock. Splitting the dock road at its stops does the same.

85% · 60-70%

Crane-use thresholds (Okpo · Yantai): past them, capacity drops sharply.

≈ 88%

Okpo's crane use from public output figures (78–97%), at the threshold.

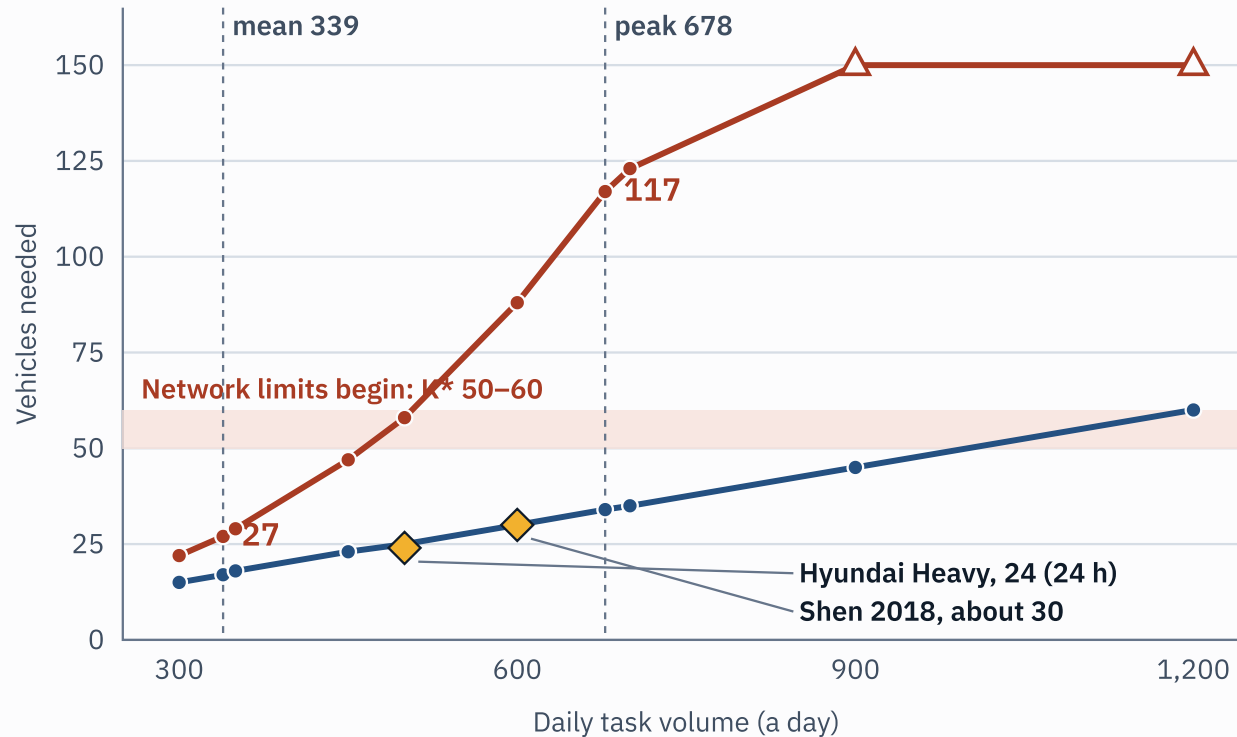
Later threshold

Waiting off the road, or splitting the dock road, lets more vehicles work at once.

What the corrections showed: vehicles needed, and where it binds

16-h day

24-h day



■ Free flow ■ Whole-route reservation ■ Field fleets ■ Network limits begin: $K^* 50-60$

16-h day: 27 vehicles for the mean of 339, 117 for the peak of 678 (whole-route reservation)

How many vehicles

Okpo's ordinary days stay below the network-bound regime; its peak falls inside it, where extra vehicles help far less. Published field fleets (Hyundai Heavy; Shen et al.) fall between the model's two lines.

The two yards fall on either side

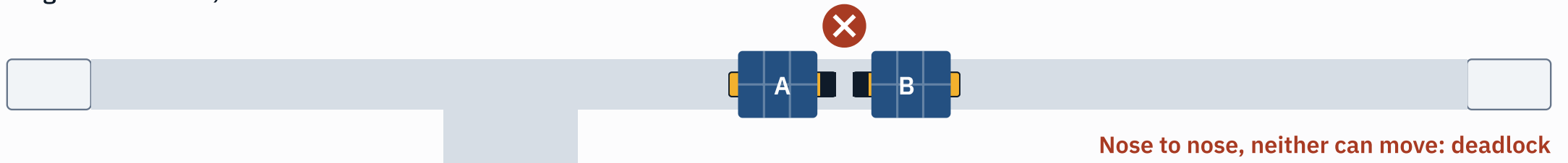
Yantai moves 23–46 blocks a day with 1–2 vehicles, free of network limits; Okpo's peak is network-bound. Hypothesis H1 yields a result either way.

Where it binds: checked on imagery, now a fixed step

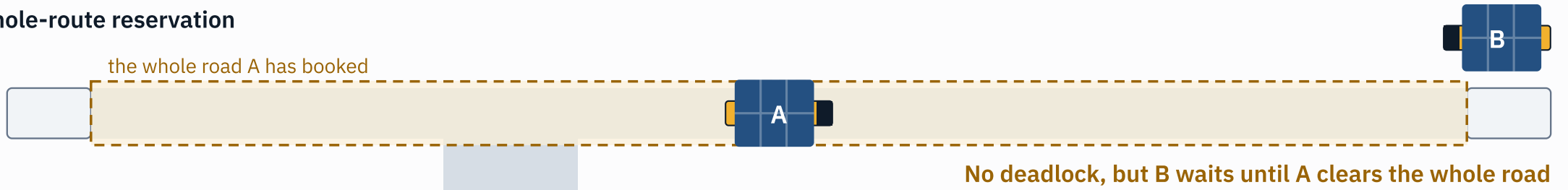
The deciding spots in both yards (Okpo's dock road 161, the doors of Yantai's building 014) sat on cautious modelling choices and changed after an imagery check. Every yard now gets its top three shadow-price spots checked: minimal human checking.

On a single lane: how deadlock arises, and how rules avoid it

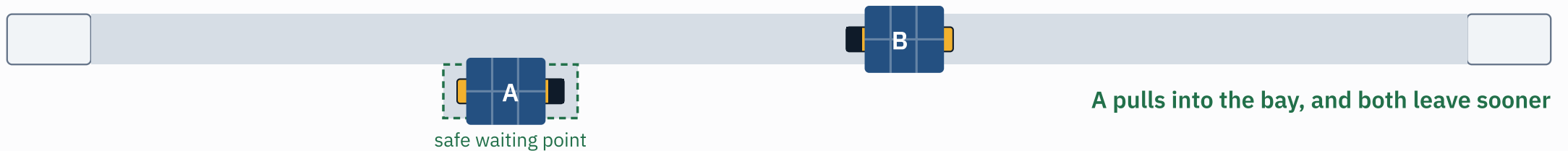
① One segment at a time, no check



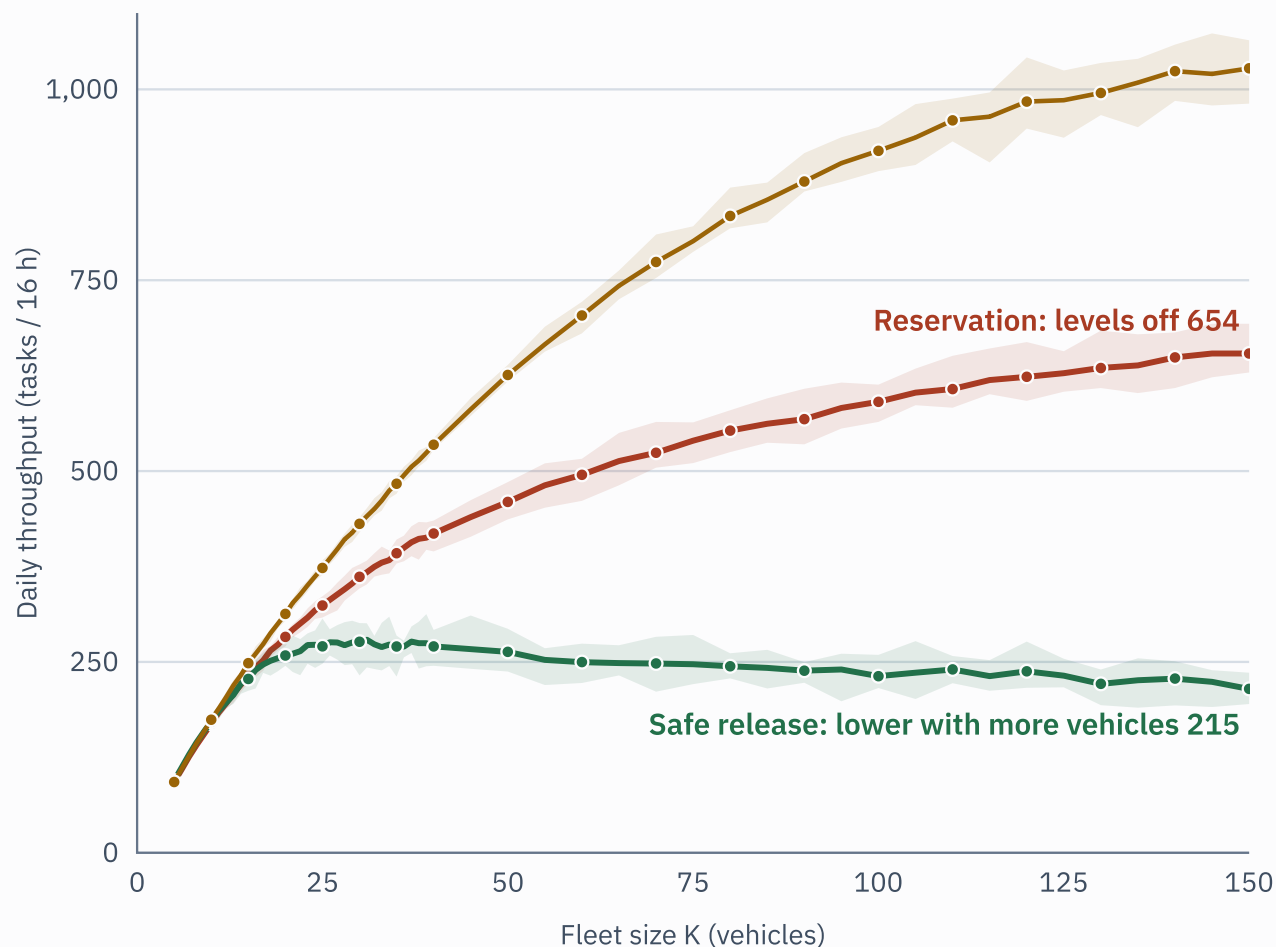
② Whole-route reservation



③ Book only to the next safe waiting point (theorem T5)



Under way: a smarter scheduling rule that still never jams (T27)



■ Whole-route reservation ■ Safe release — Segment request (reference)

Data: thesis-core a882742 (safe release on the T22 main scenario, the same source as the thesis overview's signature figure); saturated demand, mean of 10 seeds. Task board T27 (dispatched 10 Oct 2026, set by the author to precede T25); T22 review, point 4.

Why

T22's safe release is reliable but over-cautious: from 20 vehicles on it falls below reservation, and it drops as the fleet grows; about two thirds of the waiting comes from the rule refusing a release.

What

Two more tests, releasing if either passes: look a few steps ahead to see whether the move goes through; or book only to the next safe waiting point (③ on the previous slide). Step zero switches to the new map, where Okpo has 3 floating docks and new branch roads.

Why it goes first

It sets the bottom of the interval; if the bottom moves, so does the line between improving scheduling and changing roads. So T27 comes before the retrofit simulations of T25.

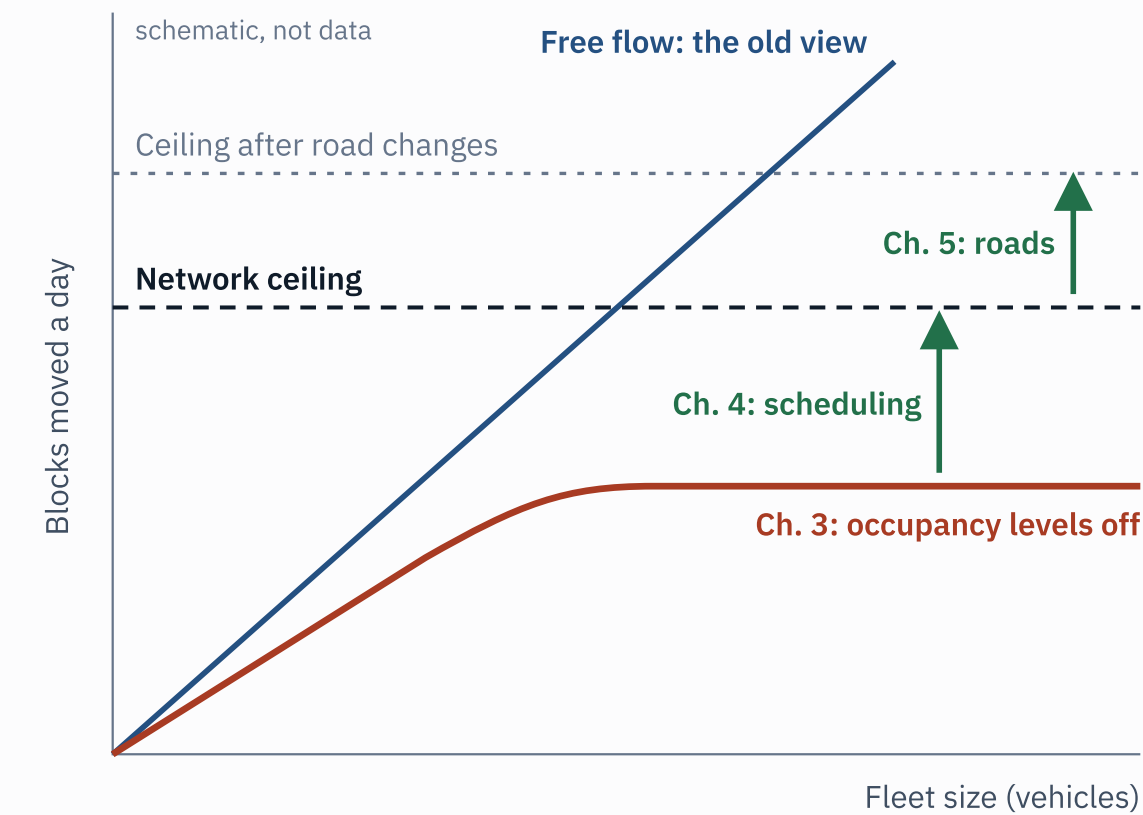
running Also chapter 4's first rule comparison (traffic orchestration).

Next: vehicles or roads, more yards, the theorems and the first paper

WORK	WHAT IT ANSWERS	WHERE IT GOES
Retrofit simulations (T25)	Road splitting, holding bays at the dock, passing bays, widening: how many vehicles each is worth, against the shadow-price prediction	Chapter 5: the first exchange-rate table
Floating docks, finite holding (T26)	Giant blocks carried by several vehicles together; with limited holding space, from what fleet size extra vehicles start to hurt	Chapter 5 (H7); chapter 3 (H2)
More yards	20 yards on the list already have maps and base images; compute the first group's bounds and test whether the map alone predicts the results	Chapter 6: testing across yards
The author: theorems	Rewrite the bound proofs (T1', T1''); prove that free-flow error can be arbitrarily large (T3) and that the exchange rate equals the shadow price (T6); settle the H1 wording after T27	The theoretical core of chapter 3
The first paper	Model and bounds, the capacity interval, the dock mouth, the first exchange-rate table; submitted within 3–4 months	First target: Transportmetrica A or Ocean Engineering

Sources: task board T25 and T26 (10 Oct 2026); shipyard list (10 Oct 2026); manifesto v1 §4; the author's review and guidance §4.2 and §4.4; direction document §12.

One figure ties the thesis together



ON THE FIGURE

IN THE THESIS

The straight line

Free flow, the old view (chapter 1)

The curve that levels off

Occupancy: where capacity comes from (chapters 2–3)

Pushing the curve up

Better scheduling (chapter 4)

Raising the ceiling

Roads or vehicles (chapter 5)

One figure per yard

Predicting from the map (chapter 6)

[Open the thesis overview →](#)

[Flag experiment and capacity interval →](#)

In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.